SOP-05: Phenol / Guanidinium Nucleic Acid Extraction Reagents (PHS)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Rooms 209 and 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	________________________________ Date: _____________
This SOP was created by (if not PI)	________________________________ Date: _____________
Emergency contact	9-1-1 (campus phone) | EH&S 206-543-7262 (M-F 8-5) | 206-685-UWPD (8973) after hours
Date of last review	_______________ Next review due: _______________ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
TRIzol LS Reagent (10296) - phenol + guanidine thiocyanate	108-95-2 / 593-84-0
TRI Reagent (R2050-1-200)	108-95-2 / 593-84-0
TRI-Reagent (TR118)	108-95-2 / 593-84-0
RNAzol RT (RB192)	108-95-2 / 593-84-0
DNAzol, DNAzol ES, DNAzol BD (DN-127, DN-128)	guanidine-detergent, proprietary
Quick-DNA Miniprep Plus Kit lysis/binding buffers (D4068)	guanidine HCl / isopropanol, proprietary
Quick-DNA/RNA Miniprep Plus Kit (D7003)	guanidine salts, proprietary
DNA/RNA Shield (R1100-250)	proprietary guanidine-based
Pico Methyl-Seq Library Prep Kit conversion reagents (D5456)	bisulfite-based, proprietary

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Skull and crossbones / Health hazard (acute toxicity, mutagenicity, STOT); Corrosion (skin corrosion Cat 1B, eye damage Cat 1); Exclamation mark

- PARTICULARLY HAZARDOUS SUBSTANCE. TRIzol LS Reagent is classified Acute toxicity Category 3 by all three routes and Germ cell mutagenicity Category 2, and phenol is a substance with high acute toxicity that is rapidly absorbed through intact skin in amounts that can be fatal. Sections 9-11 of this SOP are mandatory and apply to every reagent covered here.
- Stock chemicals: phenol/guanidinium monophase reagents (TRIzol LS, TRI Reagent, RNAzol RT) in 100-300 mL bottles; guanidine-based lysis reagents without phenol (DNAzol, DNA/RNA Shield, Quick-DNA and Quick-DNA/RNA kit buffers); bisulfite conversion reagents (Pico Methyl-Seq). Intermediates: tissue and cell lysates in these reagents; aqueous, interphase and organic phases after chloroform addition; column flow-through containing guanidine salts and isopropanol or ethanol. Wastes: organic phase and interphase (managed jointly with SOP-03), guanidinium-containing flow-through, and contaminated solid debris.
- The two dominant chemical hazards are phenol and guanidinium thiocyanate. Phenol is corrosive, is absorbed through skin extremely rapidly, and produces a characteristic painless white burn because it is a local anesthetic - a person can sustain a serious dermal exposure without feeling it. Guanidinium thiocyanate releases HIGHLY TOXIC HYDROGEN CYANIDE GAS on contact with acid or bleach; this is the single most important incompatibility in this SOP.
- Routes of exposure: skin contact and absorption (primary for phenol), inhalation of vapor or aerosol, eye contact, ingestion.
- How exposure might occur: pipetting reagent onto tissue, homogenizing or bead-beating tissue in reagent, vortexing, opening tubes after centrifugation, pipetting the aqueous phase, handling spin columns and collection tubes containing guanidinium flow-through, and consolidating waste.
- Target organs: skin, eyes, liver, kidney, central nervous system, cardiovascular system, respiratory tract. Guanidinium/cyanide exposure targets cellular respiration systemically.
- Signs and symptoms of exposure: phenol - a white, wrinkled, initially PAINLESS area of skin that later becomes a deep burn; systemic absorption causes headache, dizziness, weakness, sweating, dark urine, arrhythmia, seizures and collapse, sometimes within minutes of a modest skin exposure. Guanidinium/cyanide - headache, dizziness, confusion, rapid breathing followed by slow gasping breathing, bitter almond odor (not reliably detectable by everyone), collapse. Any suspected cyanide exposure is an immediate 9-1-1 emergency.
- Expected by-products of this process: chloroform is added to the monophase reagents in the phase-separation step (see SOP-03 - both SOPs apply simultaneously to that step); isopropanol and ethanol washes follow (SOP-01); acidified or bleached guanidinium waste would generate hydrogen cyanide, which is why this is prohibited.
- GHS classification (from MyChem / SDS): TRIzol LS - Acute toxicity Category 3 (oral, dermal, inhalation); Germ cell mutagenicity Category 2; Serious eye damage Category 1; Skin corrosion Category 1B; STOT-repeated Category 2; STOT-single Category 3. TRI Reagent / TRI-Reagent / RNAzol RT - Acute toxicity Category 4 (oral, dermal, inhalation); Skin corrosion Category 1B; Serious eye damage Category 1; STOT-repeated Category 2. DNAzol - Acute toxicity Category 4 (oral, dermal); Skin corrosion Category 1; Serious eye damage Category 1. Quick-DNA Miniprep Plus - Flammable liquids Category 2; Acute toxicity Category 4 (all routes); Skin corrosion Category 1C; Serious eye damage Category 1; STOT-single Category 3. DNA/RNA Shield - Acute toxicity Category 4 (oral); eye and skin irritation Category 2.
- Occupational exposure limits: phenol WISHA/OSHA PEL 5 ppm (19 mg/m3) 8-hr TWA, with a skin notation indicating that dermal absorption contributes significantly to total dose. There is no PEL for guanidinium thiocyanate; hydrogen cyanide has a 4.7 ppm STEL with a skin notation.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- All handling of phenol-containing reagents (TRIzol LS, TRI Reagent, RNAzol RT) - dispensing, adding to samples, homogenizing, opening tubes, phase separation, and waste transfer - is performed inside the certified chemical fume hood in FTR 209 with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S; verify current certification and airflow before starting.
- Non-phenol guanidine reagents (DNAzol, DNA/RNA Shield, Quick-DNA and Quick-DNA/RNA kit buffers) may be used at the bench for closed-tube, microliter-scale steps, but all initial dispensing from stock bottles, all lysis of tissue, and all bead-beating or homogenization is performed in the fume hood.
- Bead-beating and mechanical homogenization generate aerosol. Use sealed, O-ring or screw-cap tubes, run the homogenizer inside the fume hood, and allow tubes to settle for 1 minute before opening them - inside the hood.
- Tubes are transported between the hood and the centrifuge capped, in a closed, leak-proof secondary container, and are reopened only inside the hood.
- A disposable absorbent bench liner is used for all phenol reagent work and discarded as hazardous solid waste.
- The nearest eyewash and safety shower are located in FTR 209 and 213 and are inspected and flushed weekly. Because phenol burns require immediate, prolonged flushing and polyethylene glycol treatment, the path to the shower is kept clear at all times and a bottle of PEG 300/400 (or 70% PEG 400 in ethanol) is kept in the fume hood area for phenol skin decontamination.
- A spill kit suitable for corrosive organic liquids is maintained in FTR 209.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because these reagents are corrosive and toxic by skin contact and absorption, additional protective clothing is REQUIRED where splashes or skin contact are foreseeable: disposable chemically resistant sleeves for all phenol reagent work, plus a chemically resistant apron and face shield for any transfer of more than 50 mL and for all spill cleanup.
- Any lab coat contaminated with a phenol reagent is removed immediately, the underlying skin flushed and inspected, and the coat bagged as hazardous waste rather than laundered.

Hand protection

- Hand protection IS required for the activities described in this SOP, and glove selection is critical because phenol permeates thin nitrile rapidly.
- Required gloves for phenol-containing reagents: DOUBLE gloves - an inner 4-6 mil nitrile glove plus an outer glove of butyl rubber or laminate film (North Silver Shield / Ansell AlphaTec 02-100). Where dexterity requires it, doubled 8 mil nitrile is acceptable for microliter-scale, incidental-contact pipetting inside the hood ONLY, with the outer glove changed immediately on any contact and at least every 15 minutes.
- Required gloves for non-phenol guanidine reagents (DNAzol, DNA/RNA Shield, kit buffers): 6-8 mil nitrile, changed on any contact.

- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with phenol and with guanidinium thiocyanate. Gloves must be inspected prior to use, including a check for pinholes.
- Any glove that contacts a phenol reagent is removed immediately, and the hand is washed and inspected - phenol burns are painless and can be missed.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these reagents. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all work with phenol-containing reagents, including microliter pipetting, because these reagents cause irreversible eye damage. Safety glasses with side shields are acceptable for closed-tube handling of non-phenol kit buffers. A face shield over goggles is required for transfers of more than 50 mL and for spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided phenol reagent work is confined to the certified chemical fume hood as written.
- Phenol-containing reagents are NEVER permitted to be used outside of the chemical fume hood. Non-phenol guanidine kit buffers may be used outside the fume hood for closed-tube microliter steps and spin-column steps only; the stock bottles are opened only in the hood.
- No lab member is currently enrolled in the EH&S Respiratory Protection Program for these reagents. Respiratory protection may be needed if a dust, aerosol or vapor hazard is present and work is conducted outside of the fume hood; if any procedure may pose such a hazard, it should be eliminated or strictly isolated.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit or Occupational Exposure Limit has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator; enrollment includes medical evaluation, training and fit testing. Where air-purifying respirators are appropriate, use a full-face respirator with organic vapor cartridges as a backup to engineering controls.

Section 4 - Special Handling and Storage Requirements

- ABSOLUTE PROHIBITION: never add acid, acidic buffer, or bleach (sodium hypochlorite) to any guanidinium-containing reagent, lysate, flow-through or waste. This generates hydrogen cyanide gas and has caused fatalities in laboratories. Bleach must not be used to decontaminate surfaces or spills involving these reagents.
- Quantity limits: reagents are dispensed in volumes of no more than 50 mL at one time; routine work uses 0.3-1.0 mL per sample. Any procedure requiring more than 100 mL of a phenol reagent in a single operation requires prior written PI approval.
- Work with phenol reagents only when at least one other person is present in or immediately adjacent to the room. Unattended operations with these reagents are prohibited.
- A 'PHENOL / GUANIDINIUM REAGENTS IN USE - PARTICULARLY HAZARDOUS SUBSTANCE' sign is posted on the fume hood sash while work is in progress.
- All working aliquots, lysate tubes and waste containers must be labeled with the full reagent name (not just 'TRIzol' or 'lysate'), the hazard words 'TOXIC - CORROSIVE - CYANIDE HAZARD IF ACIDIFIED', the date and the preparer's initials.

- Storage: phenol reagents are stored in their original amber bottles in a chemically resistant secondary containment tray in the refrigerator or the dedicated toxics cabinet in FTR 209 as directed by the manufacturer, tightly closed, protected from light, segregated from acids, bases, oxidizers and bleach. Guanidine kit buffers are stored at room temperature in their kit boxes with secondary containment. Nothing in this group is stored above eye level.
- Transport: bottles are carried in a bottle carrier; tubes are carried in closed, leak-proof secondary containers. Reagents may be carried between FTR 209 and 213 in secondary containment but are opened only in the FTR 209 fume hood.
- Best practices to minimize accidents: pre-stage all consumables and the waste container inside the hood before opening any reagent bottle; work over an absorbent liner; keep the waste container inside the hood; never carry an open tube of lysate.
- Clean the chemical fume hood upon completion of tasks with a detergent solution followed by clean water and then 70% ethanol; wipe dry. DO NOT use bleach.
- Clean all contaminated surfaces with detergent solution followed by clean water and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, homogenizer probes and other surfaces contaminated with these reagents must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- NEVER use bleach on a spill of any guanidinium-containing reagent - hydrogen cyanide will be generated. Never use an acidic cleaner.
- Small spill - less than 25 mL of a phenol reagent contained INSIDE the operating fume hood, or less than 25 mL of a non-phenol guanidine buffer at the bench: may be handled by trained lab staff. Wearing goggles, face shield, butyl or laminate gloves over nitrile, lab coat with apron and sleeves, cover the spill with universal absorbent pads from the spill kit, work from the outside inward, and place used absorbent into a screw-top waste container.
- Any phenol reagent spill greater than 25 mL, ANY phenol reagent spill outside the fume hood, or any spill in which a person may have been exposed: DO NOT clean up. Evacuate the room, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S at 206-543-0467 during business hours or via 206-685-UWPD (8973) after hours.
- Liquid spills are the expected form. Dried guanidinium salt residue around bottle necks is dampened with water and wiped up, never brushed dry.
- PPE required for cleanup: chemical splash goggles AND face shield, butyl or laminate gloves over 6-8 mil nitrile, buttoned lab coat with chemically resistant apron and disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CHEMICAL SPILL - NO BLEACH' on the door for any spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the spill area thoroughly with detergent solution followed by clean water, then 70% ethanol, and dry. Repeat the detergent wash for phenol reagents.

- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, place in a screw-top container, label with a UW Hazardous Waste Label listing the reagent, 'phenol', 'guanidinium thiocyanate' and 'spill debris', and submit a collection request to EH&S promptly.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure to a PHENOL-containing reagent is a medical emergency even when it does not hurt. Immediately remove contaminated gloves and clothing and flush the skin with copious running water for at least 15 minutes using the nearest sink or safety shower. If low-molecular-weight polyethylene glycol (PEG 300/400, or 70% PEG 400 in ethanol) is available in the lab, swab the affected skin repeatedly with PEG-soaked gauze AFTER the initial water flush and continue until medical help arrives; PEG removes residual phenol that water alone leaves behind. CALL 9-1-1 for any phenol skin contact larger than a small splash, or any contact at all if the skin turns white. Do not use ethanol alone.
- Skin exposure to a non-phenol guanidine reagent: use the nearest safety shower or sink for 15 minutes; stay under the shower and remove contaminated clothing; use a clean lab coat or spare clothing for cover-up; seek medical evaluation for any burn or persistent irritation.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open, then go immediately to the Emergency Department. These reagents cause irreversible eye damage.
- Inhalation exposure: move out of the contaminated area to fresh air; get medical help. If the exposure involved acidified or bleached guanidinium material, treat it as a CYANIDE exposure - call 9-1-1 immediately, state 'possible cyanide exposure', do not enter the area to rescue without appropriate protection, and tell responders that a cyanide antidote kit may be needed.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1 immediately.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek immediate medical evaluation because subcutaneous phenol is rapidly absorbed.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (name the reagent AND state that it contains phenol and/or guanidinium thiocyanate), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused phenol/guanidinium reagent; (b) organic phase and interphase after chloroform addition - this is a HALOGENATED organic waste and is collected with the SOP-03 chloroform waste; (c) aqueous phase and guanidinium-containing spin-column flow-through, which also contains isopropanol or ethanol; (d) guanidine-only buffers and DNA/RNA Shield residues; (e) contaminated solid debris (columns, collection tubes, tips, liners, gloves, absorbent).

- CRITICAL WASTE INCOMPATIBILITY: never add any acid or any bleach to a waste container that holds guanidinium-containing material. Label every such container 'CONTAINS GUANIDINIUM - DO NOT ADD ACID OR BLEACH - CYANIDE HAZARD'.
- Keep the halogenated organic waste (chloroform-containing phases) separate from the aqueous guanidinium/alcohol waste and from the non-halogenated solvent waste of SOP-01.
- Accumulate liquid waste at the point of generation - inside the fume hood - in a sturdy, chemically compatible container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage (e.g., 'guanidinium thiocyanate 25%, phenol 5%, isopropanol 40%, water 30%'), starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Do NOT pour any phenol- or guanidinium-containing liquid down the drain. No neutralization or in-lab treatment of these wastes is authorized.
- Spin columns, collection tubes, tips and other solid debris contaminated with these reagents are collected in a lined, closed, labeled solid hazardous waste container inside or immediately adjacent to the fume hood - not the regular trash.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions - autoclaving guanidinium material is specifically prohibited.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Update the MyChem inventory when a reagent container is emptied, received, or transferred.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials.

Section 7 - Protocol

- Procedure A - Tissue lysis in TRIzol LS / TRI Reagent / RNAzol RT. Inside the certified fume hood, on an absorbent liner, wearing goggles, double gloves, lab coat and sleeves: dispense reagent from the stock bottle into a labeled 15 mL or 50 mL working tube, recap and return the stock bottle. Add 0.5-1.0 mL of reagent per 50-100 mg of bivalve or finfish tissue in a screw-cap tube. Homogenize with a bead beater or rotor-stator inside the hood using sealed tubes. Let the tube stand 1 minute before opening it, inside the hood. Incubate 5 minutes at room temperature.
- Procedure B - Phase separation. Add chloroform per SOP-03, Procedure A. SOP-03 and this SOP both apply to this step; the more protective requirement governs where they differ.
- Procedure C - Column-based extraction with guanidine kit buffers (Quick-DNA, Quick-DNA/RNA, DNAzol). Dispense lysis/binding buffer from the stock bottle inside the fume hood into a labeled working reservoir. Perform sample lysis in the hood. Closed-tube pipetting, column loading, spinning and washing may be performed at the bench with nitrile gloves, goggles or safety glasses, and a lab coat. Collect all flow-through into the labeled guanidinium waste container - never pour it into a sink.
- Procedure D - Sample preservation in DNA/RNA Shield or RNAlater. Add preservative to labeled tubes inside the hood, then add tissue at the bench or in the field. Cap securely. Field use requires the same gloves and eye protection and a leak-proof secondary container.

- Procedure E - Bisulfite conversion (Pico Methyl-Seq). Follow the manufacturer's protocol; perform all reagent dispensing in the fume hood and treat conversion reagent waste as guanidinium waste. Do not acidify.
- Prohibited: adding acid or bleach to any reagent, lysate or waste; using phenol reagents outside the fume hood; open bead-beating; drain disposal; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - these reagents are not administered to animals. They are applied to tissue after euthanasia, or used to preserve field-collected samples. If any reagent is ever used on live animals, the use will be documented and approved by IACUC and this SOP revised beforehand.

IACUC protocol number (if applicable): ______________________________

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with the reagents covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each reagent, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- These reagents include a particularly hazardous substance. Written authorization from the Principal Investigator is required before any individual may work with phenol-containing reagents, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and of the SDSs; (4) documented instruction on the guanidinium-plus-acid/bleach cyanide hazard and on phenol burn first aid including the location and use of PEG 300/400.
- A worker must complete at least two supervised runs of Procedure A and B, observed and signed off by the PI or a designated experienced lab member, before performing phenol reagent work independently for the first time. One supervised run is required for Procedure C.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider and are encouraged to discuss alternative task assignment with the PI, because TRIzol LS is classified as a germ cell mutagen Category 2. Such consultation is voluntary and confidential.
- No medical examination is required for the work described because respirators are not used. A medical examination and fit testing through the EH&S Respiratory Protection Program would be required before any respirator use.

Section 10 - Decontamination

- Fume hood: at the end of every session, remove and bag the absorbent bench liner as hazardous solid waste, then wipe the hood work surface and inside of the sash with detergent solution, then clean water, then 70% ethanol; allow to dry with the hood running. DO NOT use bleach anywhere these reagents have been used.

- Equipment: wipe pipettors, racks, and the exterior of secondary containers with detergent solution then 70% ethanol. Homogenizer probes and bead-beater tube holders are rinsed inside the hood into the appropriate waste container, then washed with detergent and water and rinsed three times with deionized water.
- Centrifuge: wipe rotors, buckets and the chamber with detergent solution and 70% ethanol after runs involving these reagents; a leaked or cracked tube requires full rotor decontamination inside the hood.
- Glassware: rinse three times inside the fume hood, collecting rinses in the appropriate waste container, then wash with detergent and water and rinse three times with deionized water.
- Controlled areas: there are no glove boxes or restricted-access hoods. The designated area in Section 11 is wiped down as described above at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, deionized water, 70% ethanol, lint-free wipes, disposable absorbent bench liners, and PEG 300/400 for skin (not for surfaces). Bleach and acids are prohibited as cleaning agents in this area.

Section 11 - Designated Area

- DESIGNATED AREA: the certified chemical fume hood in FTR 209 is the designated area for all use of phenol-containing extraction reagents (TRIzol LS, TRI Reagent, TRI-Reagent, RNAzol RT) and for all opening and dispensing of guanidine-containing kit stock bottles.
- The designated area is posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - PHENOL / GUANIDINIUM - NO ACID, NO BLEACH' sign listing the PI, contact number, required PPE and the authorization requirement.
- Secondary designated bench: the molecular biology bench in FTR 209 and the bench in FTR 213 are authorized for closed-tube and spin-column steps using non-phenol guanidine kit buffers only. These benches are posted with the 'NO ACID, NO BLEACH' warning and the guanidinium waste container is kept there in secondary containment.
- Waste containers for these reagents are kept within the designated areas in secondary containment.
- No other bench, room or piece of equipment in the Roberts Lab space is authorized for open handling of phenol-containing reagents.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

Name (print)	Signature	Date